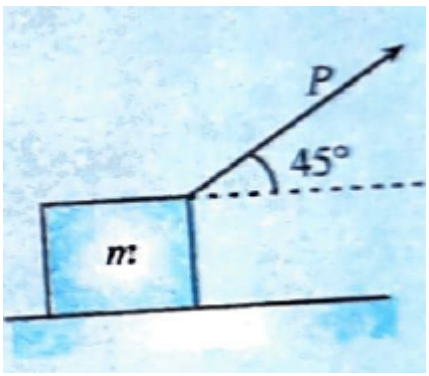
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Find the least pulling force which acting at an angle of 45° with the horizontal, will slide a body weighing $5\ \text{kg}$ along a rough horizontal surface. The coefficient of friction $\mu_s = \mu_k = 1/3$. If a force of double this is applied along the same direction, find the resulting acceleration of the block.



(A) 3/8 ms^{-2}

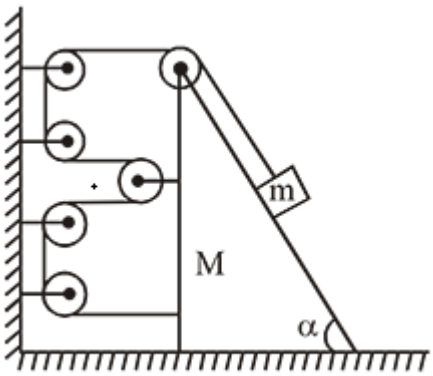
(B) 6/5 ms^{-2}

(C) 6/7 ms^{-2}

(D) 10/3 ms^{-2}

Q.2

If the acceleration of wedge in the shown arrangement is 'at towards left then at this instant acceleration of the block would be, (assume all surfaces to be frictionless)



(A) $4a$

(B) $(\sqrt{17 - 8\cos\alpha})a$

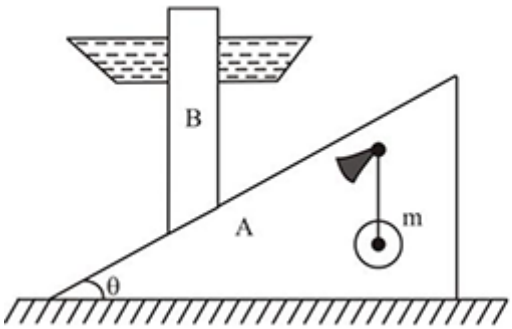
(C) $\sqrt{17a}$

(D) $\sqrt{17}\cos\frac{\alpha}{2}\text{a}$

Q.3

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In the arrangement shown, A is wedge and B a rod. The rod is constrained to move vertically down. A body of mass m is suspended by means of a string from a nail fixed to the wedge. What is the net force on the mass m if
$\alpha_B = \sqrt{3}m/s^2; \theta = 60^\circ$ and $\mathrm{m} = 2\ \mathrm{kg}$?
(Given $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) $20.1\ \mathrm{N}$

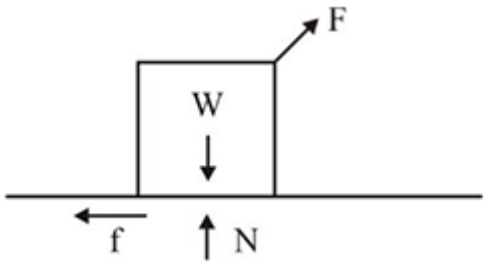
(B) $4.7\ \mathrm{N}$

(C) $19.9\ \mathrm{N}$

(D) $4.0\ \mathrm{N}$

Q.4

A person pulls a block by applying a force F and the block remains at rest. The arrows in the diagram correctly show the directions, but not necessarily the magnitudes, of the various forces on the block. Which of the followiag relations among the force maguitudes F, W, N, f is true? (f is friction and W is the weight)



(A) $\mathbf{F = f}$ and $\mathbf{N = W}$

(B) $\mathrm{F} = \mathbf{f}$ and $\mathrm{N} > \mathrm{W}$

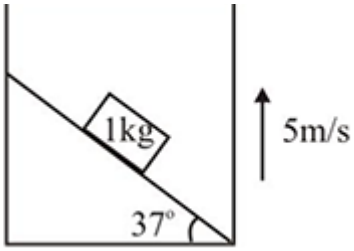
(C) $\mathrm{F} > \mathrm{f}$ and $\mathrm{N} < \mathrm{W}$

(D) $\mathrm{F} > \mathrm{f}$ and $\mathrm{N} = \boldsymbol{W}$

Q.5

A block of $1\ \mathrm{kg}$ is kept on a rough surface of an elevator moving up with constant velocity of $5\ \mathrm{m/s}$. in 10 seconds work done by normal reaction (the block does not slide on the inclined surface)
(i) from ground frame is $320\ \mathrm{J}$
(ii) is equal to work done by friction force in elevator frame
(iii)is equal to work done by friction in ground frame
Of the three statements given above, the ones that are true is given by the choice

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(A) (i)

(B) (ii), (iii)

(C) (i), (ii)

(D) only (iii)

Q.6

If $5\ \mathrm{N}$ force is acting on two bodies having mass $\mathrm{m_1\ kg}$ and $\mathrm{m_2\ kg}$ separately, then it produced acceleration $1\ \mathrm{m/s^2}$ and $\frac{1}{2}\ \mathrm{m/s^2}$ respectively. If these bodies are combined together and same force is applied on that combined body, then acceleration of combined body is

(A) $\frac{1}{3}\ \mathrm{m/s^2}$

(B) $5\ \mathrm{m/s^2}$

(C) $\frac{1}{5}\ \mathrm{m/s^2}$

(D) $\frac{2}{3}\ \mathrm{m/s^2}$

Q.7

A force produces an acceleration of $2\ \mathrm{m/s^2}$ in a body of mass $\mathrm{m_1}$ and $3\ \mathrm{m/s^2}$ in another body of mass $\mathrm{m_2}$. The acceleration produced by the force in a body of mass $\frac{m_1 m_2}{m_1 - m_2}$ is

(A) $1\ \mathrm{m/s^2}$

(B) $6\ \mathrm{m/s^2}$

(C) $5\ \mathrm{m/s^2}$

(D) $1.2\ \mathrm{m/s^2}$

Q.8

A body is moving with a speed $1\ \mathrm{m/s}$ and a force R is needed to stop it in a distance x. If the speed of the body is $3\ \mathrm{m/s}$ the force needed to stop it in the same distance x will be

(A) $1.5\ \mathrm{F}$

(B) $3\ \mathrm{F}$

(C) $6\ \mathrm{F}$

(D) $9\ \mathrm{F}$

Q.9

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A force of $16\ \mathrm{N}$ is applied on a block of mass $4\ \mathrm{kg}$ with $\mu = 0.6$. Then the force of fiction acting on the block is

(A) $20\ \mathrm{N}$

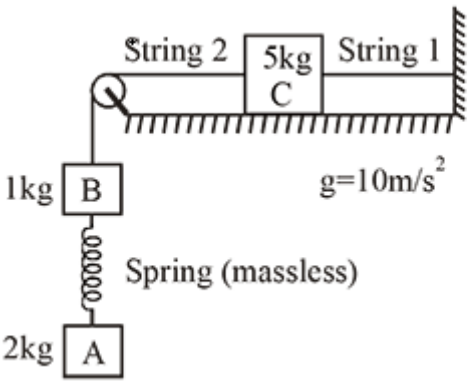
(B) $24\ \mathrm{N}$

(C) $16\ \mathrm{N}$

(D) $0\ \mathrm{N}$

Q.10

The system shown in the figure is in equilibrium and all the blocks are at rest. Assume that the masses of the strings, the pulley and the spring are negligible w.r.t. the xasses of the blocks and friction is absent. Find acceleration of the block C, just after cutting the string 1.



(A) $3\ \mathrm{m/s^2}$

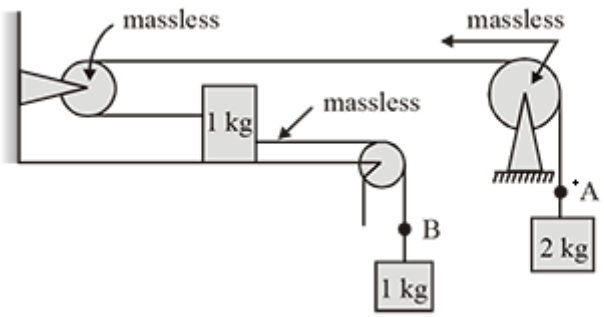
(B) $1\ \mathrm{m/s^2}$

(C) $2\ \mathrm{m/s^2}$

(D) $5\ \mathrm{m/s^2}$

Q.11

In the arrangement shown in figure, all surface are frictionless. Then the ratio of tension at point A and at point B is



(A) 1.2

(B) 2.1

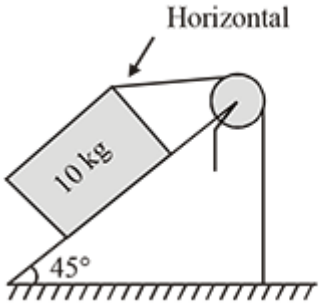
(C) 0.8

(D) 5

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Q.12

A cubical block of mass $10\ \text{kg}$ is resting on a rough inclined plane as shown in figure. The cubical block is tied by massless string which has a tension $40\ \text{N}$, The coefficient of friction between the surfaces of contact is



(A) $\frac{3}{7}$

(B) $\frac{2}{5}$

(C) $\frac{1}{4}$

(D) $\frac{2}{7}$

Q.13

A block weighing $20\ \text{N}$ rests on a rough surface. A horizontal force of $8\ \text{N}$ is required before the block start to slide, while a force of $4\ \text{N}$ keeps the block moving at a constant speed once it has started sliding. The coefficient of static friction is μ_3 and coefficient of kinetic friction is μ_k. If $\mu_s = \lambda\mu_k$ Find λ

(A) 0

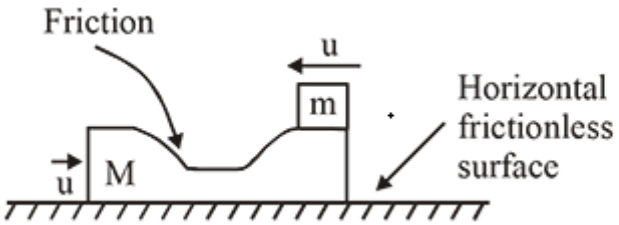
(B) 1

(C) 2

(D) 3

Q.14

A small block of mass m is placed on a bigger block of mass M, which is placed on a horizontal frictionless plane. The two blocks are given equal speeds u, but in opposite directions, as shown in the figure After sometime, it is observed that both the blocks are moving in the direction of motion of the lower block, with a velocity greater than $\frac{\text{u}}{2}$. It can be concluded that



(A) $\text{M} > 3\ \text{m}$

(B) $3\text{M} < \text{m}$

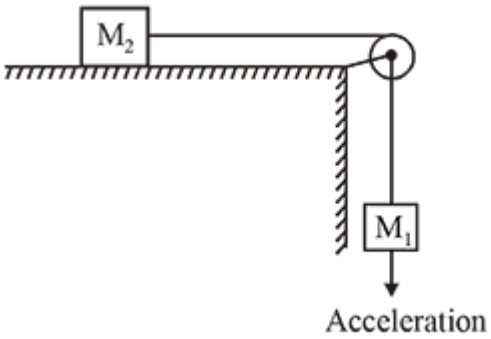
(C) $\text{m} > 2\text{M}$

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(D) none of these

Q.15

A block of mass m_2 lying on a horizontal frictionless surface is connected to a block of mass m_1 by means of string which passes over a frictionless pulley as shown in figure. If $m_1 > m_2$, the common acceleration of the masses is given by (g is the acceleration due to gravity)



(A) $\frac{m_1}{m_2}\ g$

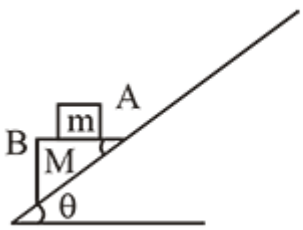
(B) $\frac{m_2}{m_1}\ g$

(C) $\frac{m_1}{(m_1+m_2)}g$

(D) $\frac{m_2}{(m_1+m_2)^8}$

Q.16

The coefficient of friction between the block A of mass m and triangular wedge B of mass M is μ. There is no friction between the wedge and the plane. If the system (block A + wedge B) is released so that there is no sliding between A and B. The inclination θ is



(A) $\theta < \tan^{-1}\mu$

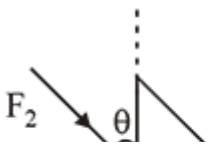
(B) $\theta < \frac{1}{2}\tan^{-1}\left(\frac{\mu m}{M}\right)$

(C) $\theta < \frac{1}{2}\cos^{-1}(2\mu)$

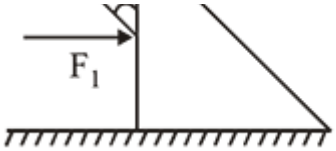
(D) $\theta < \frac{1}{2}\sin^{-1}(2\mu)$

Q.17

A wedge of mass m, lying on a rough horizontal plane, is acted upon by a horizontal force F_1 and another force F_2, inclined at an angle θ to the vertical. The block is in equilibrium, then minimum coefficient of friction between it and the surface is



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(A) $(F_2 \sin\theta) / (mg + F_2 \cos\theta)$

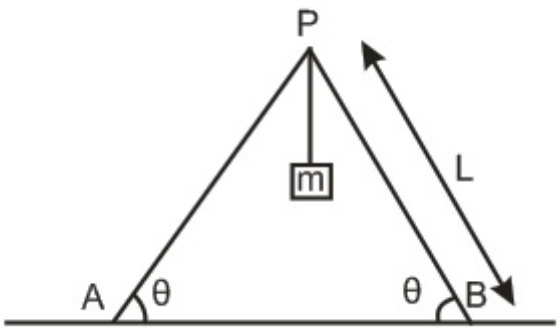
(B) $(F_1 \cos\theta + F_2) / (mg - F_2 \sin\theta)$

(C) $(F_1 + F_2 \sin\theta) / (mg + F_2 \cos\theta)$

(D) $(F_1 \sin\theta - F_2) / (mg - F_2 \cos\theta)$

Q.18

Two identical ladders are arranged as shown in the figure. Mass of the block is m and the mass of each ladder is M. The length of each of the ladder is L. The system is in equilibrium. What is the magnitude of frictional force acting at A or B?



(A) $\frac{mg}{2}\cos\theta$

(B) $\frac{Mg}{2}\cos\theta$

(C) $\frac{(M-m)}{2}g\cos\theta$

(D) $\left(\frac{M+m}{2}\right)g\cot\theta$

Q.19

A marble block of mass 2 kg lying on ice when given a velocity of 6 m s^{-1} is stopped by friction in 10 s. Then the coefficient of friction is

(A) 0.02

(B) 0.03

(C) 0.06

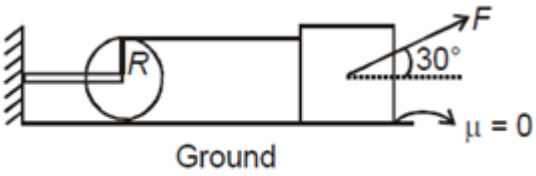
(D) 0.01

Q.20

As shown in the figure a block of mass 1 kg connected with a pulley of mass 1 kg and radius 10 cm is acted upon by a variable force of $F = kt$, as shown. Angular acceleration of pulley when block leaves the contact with

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ground is



(A) $50\sqrt{3}\ \text{rad}/\text{s}^2$

(B) $\frac{200}{\sqrt{3}}\ \text{rad}/\text{s}^2$

(C) $200\sqrt{3}\ \text{rad}/\text{s}^2$

(D) $5\sqrt{3}\ \text{rad}/\text{s}^2$

Answers & Solutions

Q.1 Answer:

10/3 ms^{-2}

Solution:

When the block is about to start sliding, the frictional force acting on block reaches to its limiting value $f = \mu_5 N$ **In vertical direction:** $N + P\sin 45^\circ = mg \Rightarrow \quad N = mg - P\sin 45^\circ$

In horizontal direction:

$P\cos 45^\circ = \mu_s N$

or $P\cos 45^\circ = \mu_s\left(mg - P\sin 45^\circ\right)$

or $\quad P = \frac{\mu_s mg}{\cos 45^\circ + \mu_s \sin 45^\circ} = \frac{25}{\sqrt{2}}\ \text{N}$

If applied force is 2P:

Friction will be kinetic nature. The block will move $f = \mu_k N.$

$N = mg - 2P\sin 45^\circ$ and $2P\cos 45^\circ - \mu_k N = ma$

$2P\cos 45^\circ - \mu_k\left(mg - 2P\sin 45^\circ\right) = ma$

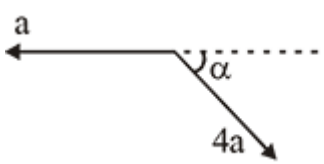
$a = \frac{2P}{m\sqrt{2}}\left(\mu_k + 1\right) - \mu_k g = \frac{10}{3}\ \text{ms}^{-2}$

Q.2 Answer:

$(\sqrt{17 - 8\cos\alpha})a$

Solution:

If the wedge moves leftward by x **then the block moves down the wedge by** $4x$ **hence acceleration of block**



$a_b = \sqrt{a^2 + (4a)^2 + 2(a)(4a)\cos(\pi - \alpha)}$

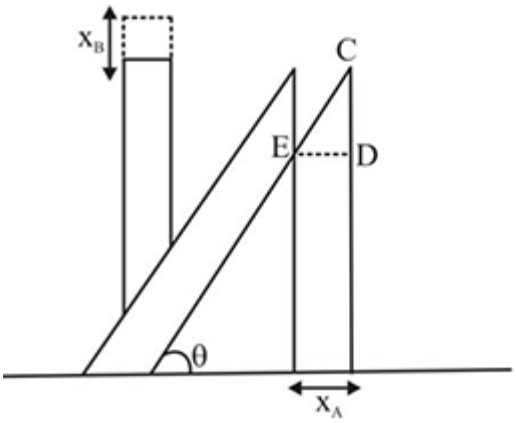
$= (\sqrt{(17 - 8\cos\alpha)})\text{a}$

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Q.3 Answer:

$20.1\ \mathrm{N}$

Solution:



Let x_A and x_B be respectively the displacement of wedge and rod as shown in figure.

In $\Delta CDE, BD = x_A$ and $CD = x_B$

$\therefore \tan\theta = \frac{\mathrm{CD}}{\mathrm{BD}} = \frac{\mathrm{x_B}}{\mathrm{x_A}} \therefore \mathrm{x_B} = \mathrm{x_A}\tan\theta$

Differentiating twice we get m

$\mathrm{x_B} = \mathrm{x_A}\tan\theta \therefore \mathrm{a_B} = \mathrm{a_A}\tan\theta$

$\therefore a_A = \frac{a_B}{\tan\theta} = \frac{\sqrt{3}}{\sqrt{3}} = 1 \therefore a_A = \mathrm{lm/s^2}$

Since the mass m is in an accelerated frame of reference it is subjected to a pseudo force in addition to gravitational force. The two forces are acting at right angles. Hence the net force on m is

$F_{nt} = \sqrt{(2\times 10)^2 + (2\times 1)^2} = \sqrt{404}\ \mathrm{N} = 20.1\ \mathrm{N}$

Q.4 Answer:

$\mathrm{F} > \mathrm{f}$ and $\mathrm{N} < \mathrm{W}$

Solution:

$\mathbf{F_a = f \Rightarrow F > f}$ **and** $\mathbf{F}_y = \mathbf{N = w \Rightarrow N < w}$

Q.5 Answer:

(i)

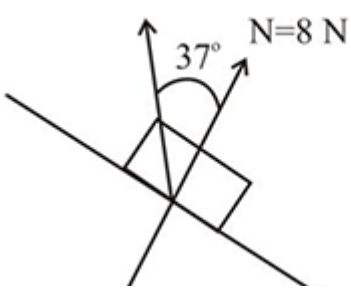
Solution:

$N\cos 37^\circ = 8\times 0.8 = 6.4\ \mathrm{N}$

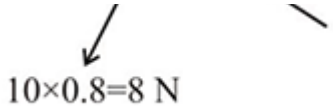
W.D. from ground trame

$= 6.4\times 50 = 320\ \mathrm{J}$

From elevator frame, displacement $= 0$



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Q.6 Answer:

$\frac{1}{3}\ \mathrm{m/s^2}$

Solution:

$a = \frac{a_1 a_2}{a_1 + a_2} \Rightarrow a = \frac{1 \times \frac{1}{2}}{1 + \frac{1}{2}} = \frac{1}{3} m/s^2$

Q.7 Answer:

$1\ \mathrm{m/s^2}$

Solution:

$\frac{1}{m} = \frac{1}{m_2} - \frac{1}{m_1}$
$\frac{F}{m} = \frac{F}{m_2} - \frac{F}{m_1}$
$a = a_2 - a_1 = 3 - 2 = 1\ \mathrm{m/s^2}$

Q.8 Answer:

$9\ \mathrm{F}$

Solution:

$F_x = \frac{1}{2} m v^2$
$F' x = \frac{1}{2} m (3v)^2$
Hence $\mathrm{v} = |\ \mathrm{m}$
$\frac{\mathbf{F'}}{\mathbf{F}} = 9$

Q.9 Answer:

$16\ \mathrm{N}$

Solution:

When F applied in lesser than limiting friction.

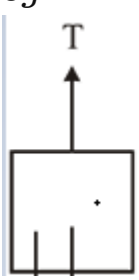
Q.10 Answer:

$5\ \mathrm{m/s^2}$

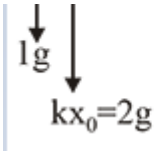
Solution:

Just after cutting the string 1 the extension of the string will remain same as it was before cutting string 'l'. So, we can take string force on block equal to ' $2\ \mathrm{g}$ **'. Now making F.B.D of block 'B'.**

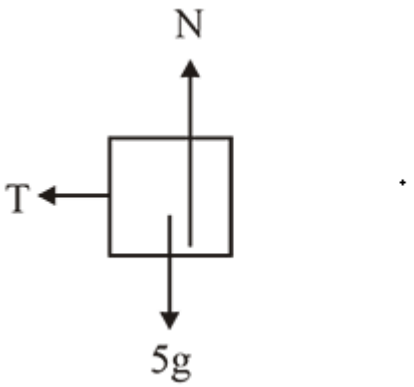
$3g - T = 1a$



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Making F.B.D of block 'c'



$T = 5a \quad \ldots (2)$

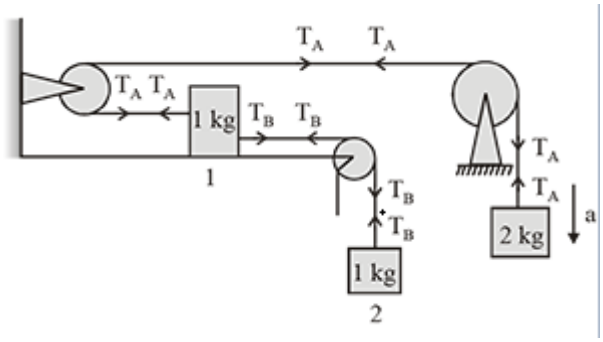
From (1) and (2)

$a = g/2 = 5\ m/s^2$

Q.11 Answer:

1.2

Solution:



For $2\ kg$

$2g - T_A = 2a \ldots .$ **(i)**

For block (1)

$\mathbf{T_A - T_B = a} \ldots \ldots$

Fox block (2)

$T_B - g = a \ldots \ldots (ii)$

from equation (i), (ii) \& (iii)

$g = 4a \Rightarrow a = \frac{g}{4}$

from equation (ij)

$T_B - g = \frac{g}{4} \Rightarrow T_B = \frac{5\,g}{4}$

from equation (i)

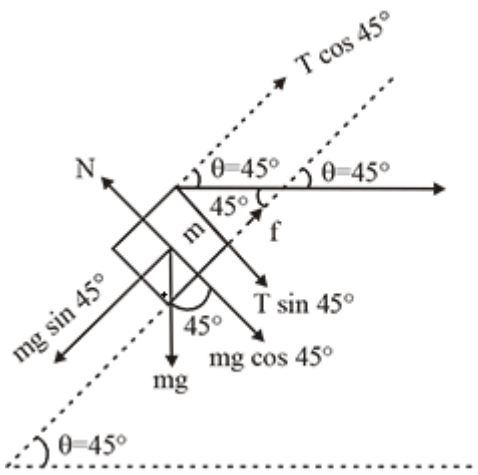
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$2g - T_A = \frac{2g}{4} = \frac{g}{2}$
$\Rightarrow T_A = \frac{3g}{2}$ ie $\frac{T_A}{T_B} = \frac{6}{5} = 1.2$

Q.12 Answer:

$\frac{3}{7}$

Solution:



$\mathrm{T}\cos 45^\circ + f = \mathrm{mg}\sin 45^\circ$
$\frac{T}{\sqrt{2}} + \mu\mathrm{N} = \frac{100}{\sqrt{2}} \Rightarrow \frac{40}{\sqrt{2}} + \mu N = \frac{100}{\sqrt{2}} \mu\mathrm{N} = \frac{60}{\sqrt{2}}$ ..... **(i)**
$N = mg\cos 45^\circ + T\sin 45^\circ$
(ii) $= \frac{100}{\sqrt{2}} + \frac{40}{\sqrt{2}} = \frac{140}{\sqrt{2}}$
\cdots
from (i) and (ii)
$\mu = \frac{\mu\mathrm{N}}{\mathrm{N}} = \frac{60}{140} = \frac{3}{7}$

Q.13 Answer:

2

Solution:

$\mu_\mathrm{s}(20) = 8 \Rightarrow \mu_\mathrm{s} = 0.4$
$\mu_\mathrm{k}(20) = 4 \Rightarrow \mu_\mathrm{k} = 0.2$
$\mu_\mathrm{s} = \lambda\mu_\mathrm{k}$
Hence $\lambda = 2$

Q.14 Answer:

$\mathrm{M} > 3\ \mathrm{m}$

Solution:

initially both are moving in opposite direction and after some time both are moving in same direction which is possible if $\mathrm{P}_f > \mathrm{P}_t$
$\mathbf{m}\frac{\mathbf{u}}{2} + \mathbf{M}\frac{\mathbf{u}}{2} > \mathrm{Mu} - \mathbf{mu}$
$(\mathbf{m} + \mathbf{M})\frac{\mathbf{u}}{2} > (\mathbf{M} - \mathbf{m})\mathbf{u}$

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$M > 3\ m$

Q.15 Answer:

$\frac{m_1}{(m_1+m_2)}g$

Solution:

masses m_1 and m_2 are interchanged. Therefore, for masses m_1 and m_2, we have $T = m_2 a$ ..... (i);

$m_1 g - T = m_1 a$ .....(ii)

Adding the two equations, we get a $\frac{m_1 g}{(m_1+m_2)}$

Q.16 Answer:

$\theta < \tan^{-1} \mu$

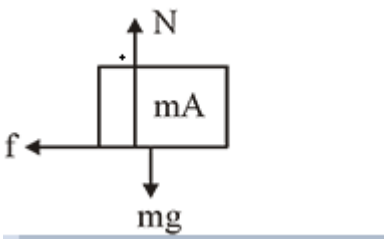
Solution:

Acceleration of the system $(A+B)$ down the plane $= g \sin\theta$

$f = m(g \sin\theta \cdot \cos\theta) = \frac{mg}{2} \sin 2\theta$

for no sliding between A and B

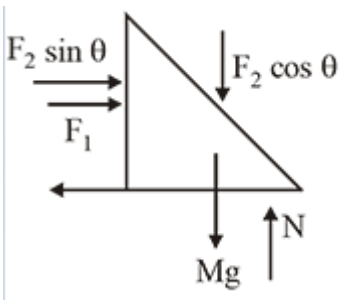
$f < \mu N \Rightarrow \frac{mg}{2} \sin 2\theta < \mu\, mg\, \theta < \frac{1}{2} \sin^{-1}(2\mu)$



Q.17 Answer:

$(F_1 + F_2 \sin\theta) / (mg + F_2 \cos\theta)$

Solution:



$f = \mu N; (F_1 + F_2 \sin\theta) = \mu\,(mg + F_2 \cos\theta)$

$\mu = \frac{F_1 + F_2 \sin\theta}{mg + F_2 \cos\theta}$

Q.18 Answer:

$\frac{M+m}{2} g \cot\theta$

Solution:

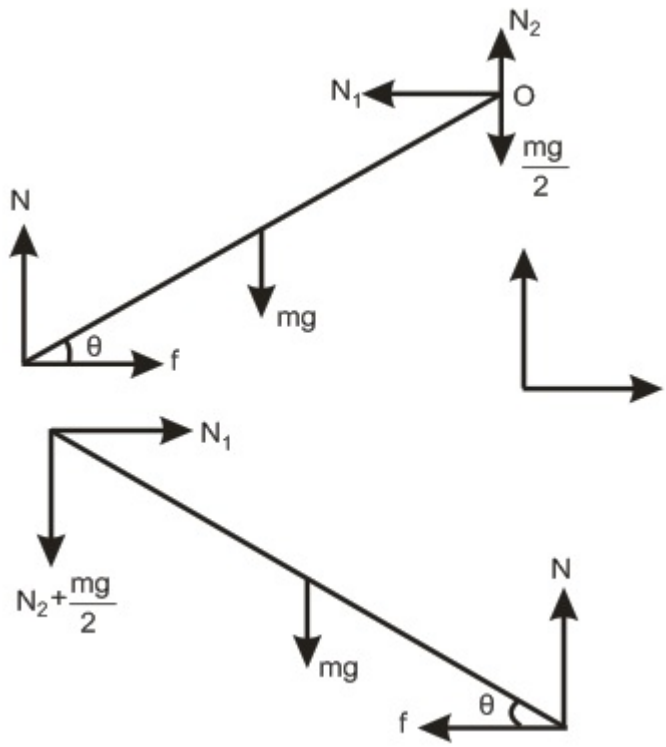
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Drawing force diagrams of the rod, we have

This is the equilibrium of coplanar forces, hence using the equations

$\sum F_x = 0, \sum F_y = 0$ **and**

Net moment about point O **is zero.**



we have the equations,

$N + N_2 = Mg + \frac{mg}{2}$**... (i)**

$N_1 = f$ **...(ii)**

$N = N_2 + Mg + \frac{mg}{2}$ **... (iii)**

and $Mg\frac{L}{2} \cos\theta + fL\sin\theta = NL\cos\theta$ **... (iv)**

Solving these four equations, we have

$f = \frac{M+m}{2} g \cot\theta$

Q.19 Answer:

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0.06

Solution:

Let coefficient of friction be μ**, and then retardation will be** μg**.**

From equation of motion, $v = u + at$

$\Rightarrow \quad 0 = 6 - \mu \ g \times 10$

$\Rightarrow \quad \mu = \frac{6}{100} = 0.06$

Q.20 Answer:

$200\sqrt{3}$ rad / s^2

Solution:

Resolving F

$F \quad \cos 30° - T = ma$ **.... (i)**

$F \quad \sin 30° + N = mg$ **.... (ii)**

For the time it leaves the contact

$F \quad \sin 30° = mg$

$F = 2 \quad \text{mg}$

$T.R. = I\alpha$

$2 \quad mg \quad \cos 30 - T = mr\alpha$

$T = \sqrt{3} \quad \text{mg} - mr\alpha$

$T.R. = I.\alpha$

$\{\sqrt{3} \times 1 \times 10 - 1 \times 0.1 \quad \alpha\} \times 0.1 = \frac{1 \times 0.01}{2} \times \alpha$

$\alpha = 200\sqrt{3}$ rad / s^2

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